

Topic: Future of IoT in Smart Cities

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1: Title Page & Abstract

Future of IoT in Smart Cities: A Case Study

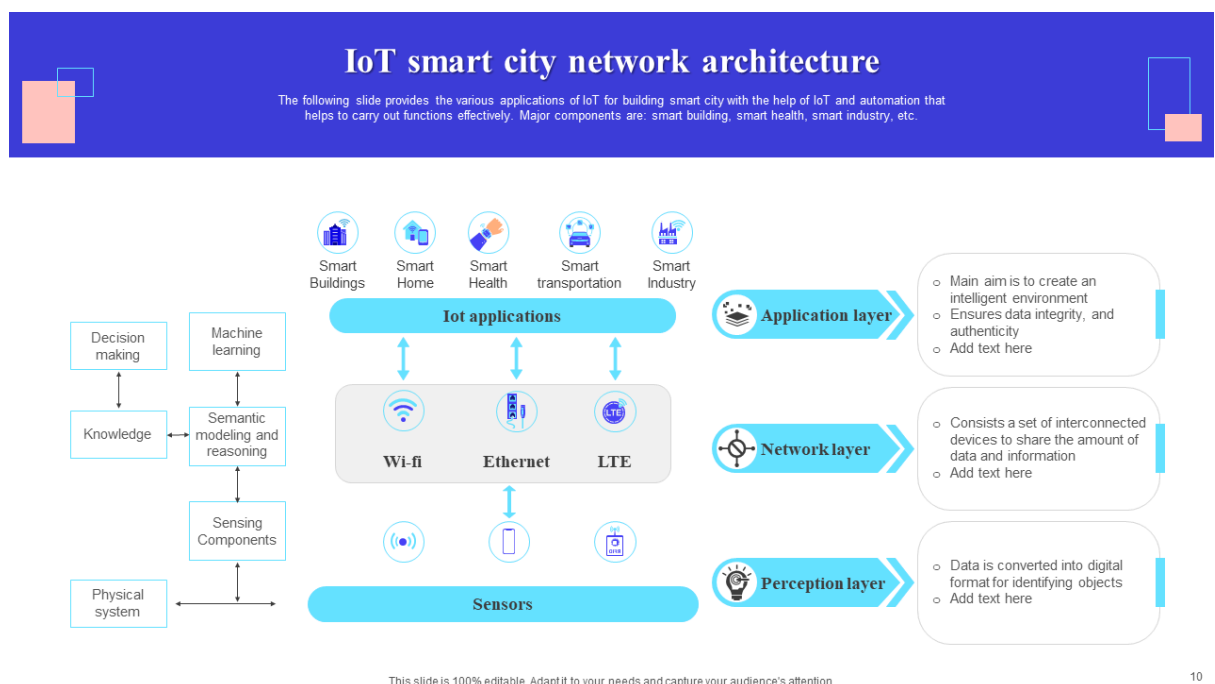
Abstract: This case study explores how the Internet of Things is transforming urban life. By 2030, 70% of the world population will live in cities. IoT provides solutions for traffic, pollution, energy, and safety. This study analyzes architecture, real-world implementations like Lucknow Smart City, challenges and future scope.

2: Introduction & Architecture

1. Introduction: Smart City means using technology to improve quality of life. IoT is its backbone.

2. Four-Layer Architecture:

- **Sensing Layer:** Sensors - LDR, DHT11, MQ135 air quality, ultrasonic for parking.
- **Network Layer:** Data transfer via WiFi, LoRaWAN, 5G.
- **Data Processing Layer:** Cloud platforms like AWS IoT, Blynk, data analytics.
- **Application Layer:** User apps - Smart traffic lights, waste management dashboard.



3: Key Applications (with examples)

1. Smart Traffic Management: IoT sensors at signals count vehicles, AI adjusts green time.

Case: Barcelona reduced traffic jams by 21%.

2. Smart Waste Management: Ultrasonic sensor in dustbin alerts when full. Route optimization saves fuel.

3. Smart Energy & Street Lights: LDR based lights

4. Smart Water & Pollution Monitoring: Water level sensor prevents overflow, MQ135 sensor sends AQI on mobile app.

5. Smart Safety: CCTV + IoT panic buttons.

4: Case Study - Lucknow Smart City & Global Case

Lucknow Example: Under Smart City Mission, Hussainganj me adaptive traffic lights, Gomti Nagar me smart parking sensors, 1100+ CCTV with IoT integration.



Global Case: Singapore: World's No.1 smart city. Uses IoT for everything - from predicting bus arrival to monitoring elderly health at home.

5: Challenges & Future Scope

- Security: Hackers can hack traffic systems.
- Cost: High initial cost.
- Interoperability
- Data Privacy.

Future Scope (2025-2030):

- AI + IoT = AIoT: Predictive maintenance.
- 5G se real-time data.
- Digital Twin
- Drone based delivery monitored by IoT.

6: Conclusion & References
Conclusion: IoT is not the future, it is the present. With Task 2 like small projects, we are building a base for smart cities. India needs more students like us to work on IoT.

References:

1. McKinsey Report on Smart Cities 2023
2. www.codealpha.tech
3. India Smart City Mission - smartcities.gov.in
4. IBM Institute - IoT in Urban Development